

Pixel Art Is a Palette, Not a Downscale: Rebuilding Targets and Decoding for Tiny Sprite Generation

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Abstract

001 Text-to-image models fail at the smallest images people
002 draw on purpose: 12–24px RGBA game sprites,
003 where a handful of colours and hard alpha edges are
004 the medium. We show that the usual failure is not
005 a modelling failure but a target failure. In the sprite
006 corpus this area trains on, only 7.1% of the images are
007 natively 16px or smaller; the rest is 25–64px artwork
008 box-downscaled into the low-resolution buckets, and
009 4,888 pixel-identical duplicates leave a twin in training
010 for 16.2% of the held-out set. A model fitted to those
011 targets learns soft, fifty-colour blobs, and the evalua-
012 tion reference, drawn from the same pool, rewards ex-
013 actly that. We rebuild the targets and evaluate against
014 references that contain only real low-resolution pixel
015 art. Under this protocol the data work alone takes FD-
016 DINOv2 from 167.2 to 117.9 at 16px. We then close
017 the remaining gap where it actually lies: real sprites
018 use a median of six opaque colours while every sam-
019 pling rule we tested emits more than fifty. Projecting
020 the predicted $\hat{x}_0$ onto a per-sprite k -colour palette dur-
021 ing the late denoising steps reaches 31.8 ± 0.5 — bet-
022 ter than quantising finished samples (41.6), than one
023 corpus-wide palette (120.1) and than a median filter
024 that removes the same dither (112.2). The rule is a
025 property of the recipe rather than of the network: a
026 transformer backbone that trails the UNet by 36.5 FD
027 without it lands within 0.7 with it. Applied to sprites
028 from generators we did not train, it improves native
029 FD by 36–41% while preserving 80% of the silhouette.
030 Against external systems given the same palette post-
031 process we reach 50.8 versus 134.4 for gpt-image-2 with
032 downscaling.

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